

1 Summary

Model (Run-name): model3 (test)

Iteration: 1

TAC: 57934.361890328146

Took: 2.9105513095855713s

Total Time: 2.9105513095855713s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

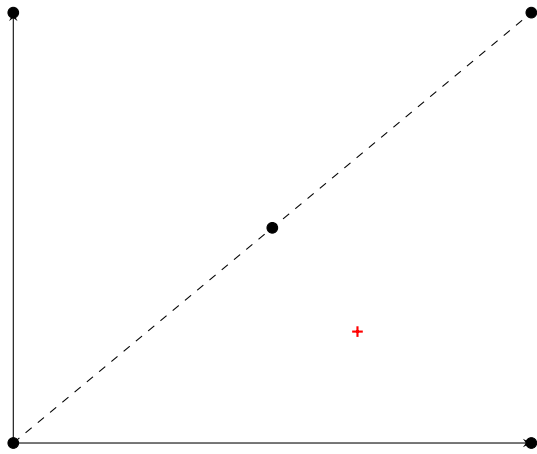
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{2,2,1}$	27.2	$q_{1,1,2}$	229.55
$z_{1,4,2}$	1.0	$dt_{2,2,2}$	22.2	$q_{1,4,2}$	357.4100000000001
$z_{2,1,1}$	1.0	$dt_{3,5,1}$	22.3	$q_{2,1,1}$	530.45
$z_{2,2,1}$	1.0	$dt_{3,5,2}$	29.056081081081047	$q_{2,2,1}$	647.20999999999999
$z_{3,5,1}$	1.0	$dt_{4,3,1}$	50.0	$q_{3,5,1}$	1544.2900000000002
$z_{4,3,1}$	1.0	$dt_{4,3,2}$	111.1	$q_{4,3,1}$	1539.7199999999998
$z_{5,4,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{5,4,1}$	1277.4399999999998
$z_{cu,3}$	1.0	$dt_{5,4,2}$	23.86871101531634	$q_{cu,3}$	839.9899999999997
$z_{cu,5}$	1.0	$dt_{cu,3}$	82.35608108108106	$q_{cu,5}$	1081.9700000000003
$A_{1,1,2}^\beta$	0.4963437290482565	$dt_{cu,5}$	86.72824858757063	$f_{1,1}^{(H)}$	160.0
$A_{1,4,2}^\beta$	0.9933272385456146	$f_{1,1,2}^H$	3.441529235382309	$f_{1,2}^{(H)}$	160.0
$A_{2,1,1}^\beta$	0.5475650289437815	$f_{1,4,2}^H$	5.358470764617692	$f_{1,3}^{(H)}$	93.3
$A_{2,2,1}^\beta$	3.338874959957603	$f_{2,1,1}^H$	4.774527452745275	$f_{2,1}^{(H)}$	248.9
$A_{3,5,1}^\beta$	5.563592903508788	$f_{2,2,1}^H$	5.825472547254725	$f_{2,2}^{(H)}$	137.8
$A_{4,3,1}^\beta$	2.4894904573733196	$f_{3,5,1}^H$	9.585909373060213	$f_{2,3}^{(H)}$	137.8
$A_{5,4,1}^\beta$	1.57118431451739	$f_{4,3,1}^H$	12.6	$f_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	1.4797875754821508	$f_{5,4,1}^H$	9.583195798949735	$f_{3,2}^{(H)}$	122.35608108108107
$A_{cu,5}^\beta$	2.114093315418625	$f_{1,1,2}^C$	7.6	$f_{3,3}^{(H)}$	122.35608108108106
$A_{1,1,2}$	2.7005882352941173	$f_{1,4,2}^C$	3.782116402116402	$f_{4,1}^{(H)}$	271.1
$A_{1,4,2}$	5.40465749281718	$f_{2,1,1}^C$	5.3044999999999995	$f_{4,2}^{(H)}$	148.9
$A_{2,1,1}$	3.3036464983028657	$f_{2,2,1}^C$	6.1	$f_{4,3}^{(H)}$	148.9
$A_{2,2,1}$	18.890814020083003	$f_{3,5,1}^C$	13.9	$f_{5,1}^{(H)}$	198.9
$A_{3,5,1}$	44.57376487924321	$f_{4,3,1}^C$	8.4	$f_{5,2}^{(H)}$	126.72824858757068
$A_{4,3,1}$	19.921388087224845	$f_{5,4,1}^C$	13.517883597883595	$f_{5,3}^{(H)}$	126.72824858757063
$A_{5,4,1}$	12.878068451030797	$RecLMTD_{1,1,2}$	0.019322966507177033	$f_{1,1}^{(C)}$	160.0
$A_{cu,3}$	10.320104112088831	$RecLMTD_{1,4,2}$	0.046934985124871535	$f_{1,2}^{(C)}$	90.20394736842101
$A_{cu,5}$	14.682070969056184	$RecLMTD_{2,1,1}$	0.013210108961198144	$f_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	69.79605263157899	$RecLMTD_{2,2,1}$	0.024809863749124014	$f_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{3,5,1}$	0.02453405781773936	$f_{2,2}^{(C)}$	115.6
$dt_{1,4,2}$	19.51300297502571	$RecLMTD_{4,3,1}$	0.010997570905191309	$f_{2,3}^{(C)}$	115.6
$dt_{1,4,3}$	11.1	$RecLMTD_{5,4,1}$	0.026805504819141977	$f_{3,1}^{(C)}$	221.1
$dt_{2,1,1}$	88.9	$RecLMTD_{cu,3}$	0.021128736072437457	$f_{3,2}^{(C)}$	37.8
$dt_{2,1,2}$	47.596052631579084	$RecLMTD_{cu,5}$	0.02204673436571182	$f_{3,3}^{(C)}$	37.8

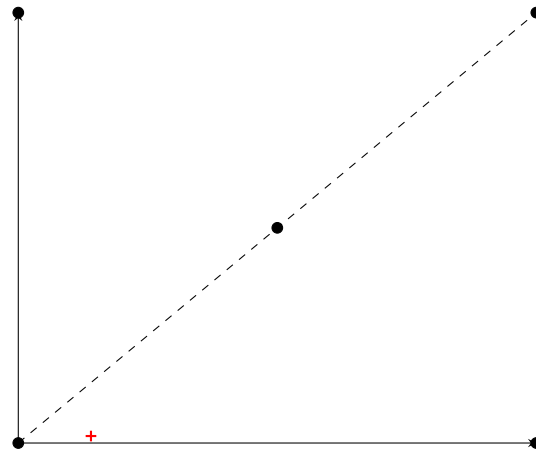
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,1}^{H,out}$	628.6576444111025
$t_{4,2}^{(C)}$	102.85953757225434	$b_{1,1,2}^{C,in}$	456.0
$t_{4,3}^{(C)}$	82.2	$b_{1,4,2}^{C,in}$	310.88996825396833
$t_{5,1}^{(C)}$	204.4	$b_{2,1,1}^{C,in}$	318.26999999999975
$t_{5,2}^{(C)}$	93.3	$b_{2,2,1}^{C,in}$	705.16
$t_{5,3}^{(C)}$	93.3	$b_{3,5,1}^{C,in}$	1296.87
$t_{1,1,2}^H$	133.91477272727275	$b_{4,3,1}^{C,in}$	317.51999999999987
$t_{1,4,2}^H$	119.38522727272726	$b_{5,4,1}^{C,in}$	1111.1700317460316
$t_{2,1,1}^H$	198.8575471698114	$b_{1,1,2}^{C,out}$	685.55
$t_{2,2,1}^H$	187.84245283018873	$b_{1,4,2}^{C,out}$	668.2999682539685
$t_{3,5,1}^H$	122.35608108108107	$b_{2,1,1}^{C,out}$	848.7199999999993
$t_{4,3,1}^H$	148.9	$b_{2,2,1}^{C,out}$	1352.3699999999997
$t_{5,4,1}^H$	126.72824858757063	$b_{3,5,1}^{C,out}$	2841.16
$t_{1,1,2}^C$	90.20394736842104	$b_{4,3,1}^{C,out}$	1857.2399999999993
$t_{1,4,2}^C$	102.85953757225434	$b_{5,4,1}^{C,out}$	2388.610031746031
$t_{2,1,1}^C$	129.7960526315789		
$t_{2,2,1}^C$	221.7		
$t_{3,5,1}^C$	204.4		
$t_{4,3,1}^C$	221.1		
$t_{5,4,1}^C$	156.04046242774564		
$b_{1,1,2}^{H,in}$	550.6446776611695		
$b_{1,4,2}^{H,in}$	857.3553223388308		
$b_{2,1,1}^{H,in}$	1188.379882988299		
$b_{2,2,1}^{H,in}$	1449.960117011701		
$b_{3,5,1}^{H,in}$	2173.12565487275		
$b_{4,3,1}^{H,in}$	3415.8599999999988		
$b_{5,4,1}^{H,in}$	1906.0976444111022		
$b_{1,1,2}^{H,out}$	321.0946776611694		
$b_{1,4,2}^{H,out}$	499.9453223388307		
$b_{2,1,1}^{H,out}$	657.9298829882995		
$b_{2,2,1}^{H,out}$	802.7501170117011		
$b_{3,5,1}^{H,out}$	628.8356548727498		
$b_{4,3,1}^{H,out}$	1876.1399999999994		

3 *RecLMTD* Tangent Points

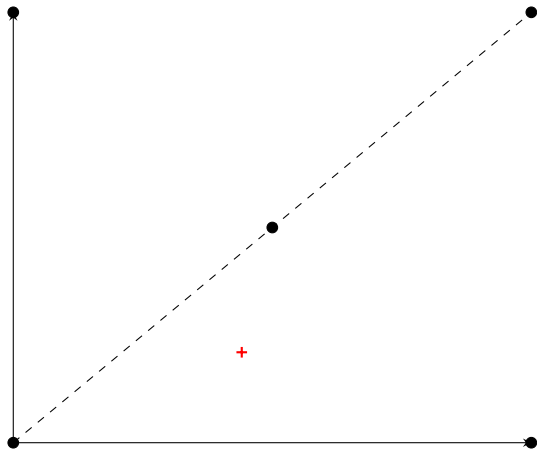
- (1, 1, 2)
 - Value: 0.019323
 - Error: 0.000953748514498
 - Relative error: 0.047036638502782
- (1, 4, 2)
 - Value: 0.046935
 - Error: 0.020120257329339
 - Relative error: 0.300054948620580
- (2, 1, 1)
 - Value: 0.013210
 - Error: 0.001915862100514
 - Relative error: 0.126660436721523
- (2, 2, 1)
 - Value: 0.024810
 - Error: 0.015815073135620
 - Relative error: 0.389294712764435
- (3, 5, 1)
 - Value: 0.024534
 - Error: 0.014636744202497
 - Relative error: 0.373664654477470
- (4, 3, 1)
 - Value: 0.010998
 - Error: 0.002069658083637
 - Relative error: 0.158385384185647
- (5, 4, 1)
 - Value: 0.026806
 - Error: 0.016626910472062
 - Relative error: 0.382822607506002
- $cu, 3$
 - Value: 0.021129
 - Error: 0.004190263559839
 - Relative error: -0.247381430451997
- $cu, 5$
 - Value: 0.022047
 - Error: 0.005592357551558
 - Relative error: -0.339870516806623



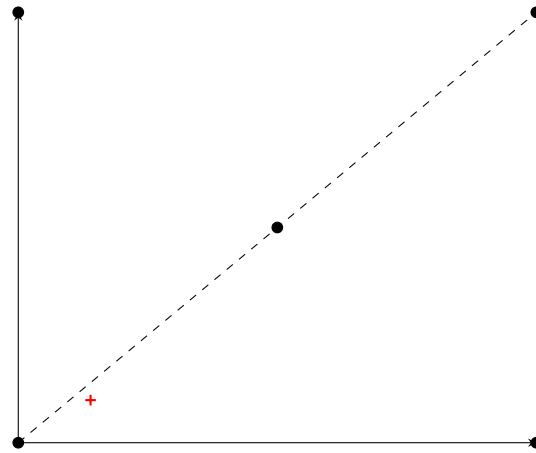
(a) *RecLMTD* - (1, 1, 2)



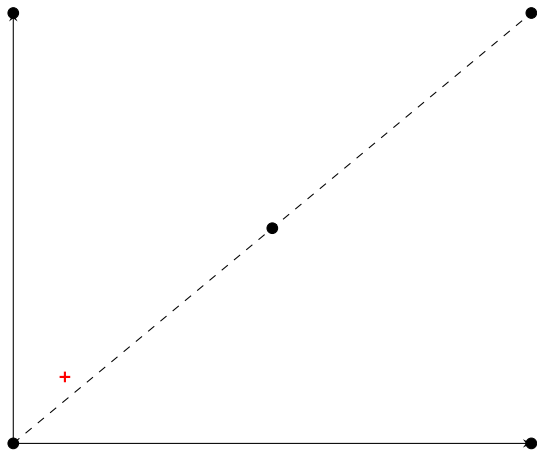
(b) *RecLMTD* - (1, 4, 2)



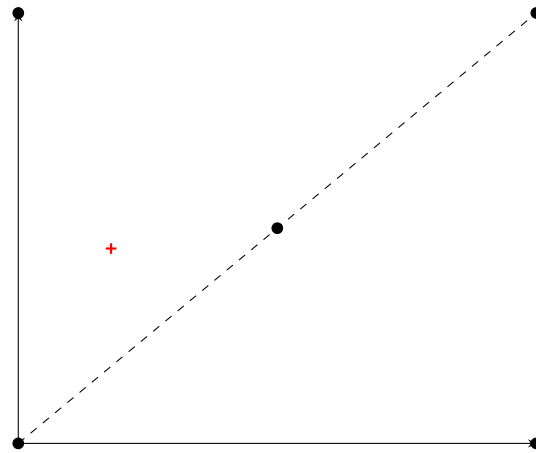
(c) *RecLMTD* - (2, 1, 1)



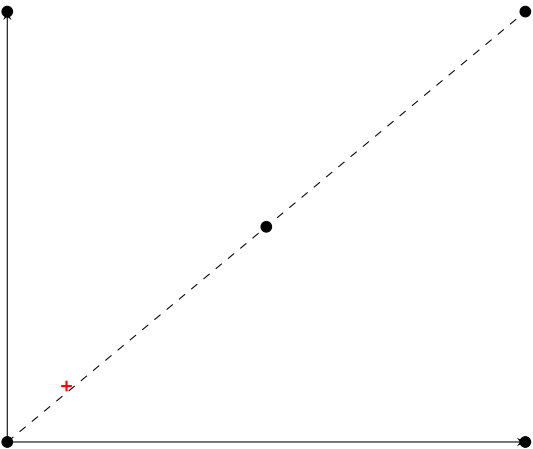
(d) *RecLMTD* - (2, 2, 1)



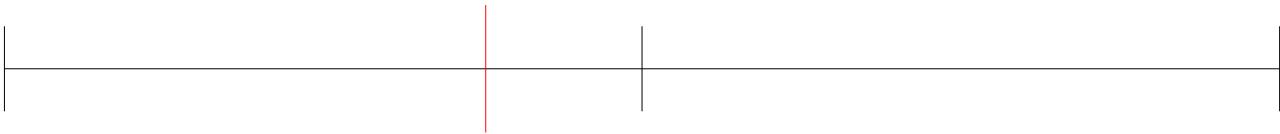
(e) *RecLMTD* - (3, 5, 1)



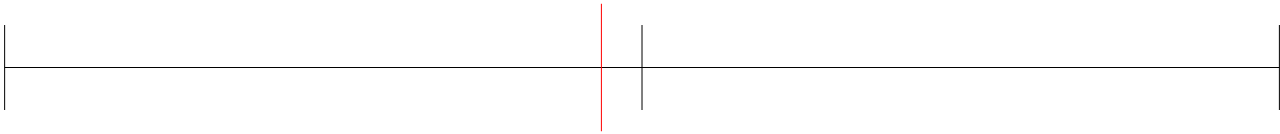
(f) *RecLMTD* - (4, 3, 1)



(g) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 550.6446776611695
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 4, 2)
 - Value: 857.3553223388308
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1188.379882988299
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2173.12565487275
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 3415.8599999999988
 - Absolute error: 1.3642420526593924e-12
 - Relative error: -3.9938465061782166e-16
- (5, 4, 1)
 - Value: 1906.0976444111022
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 321.0946776611694
 - Absolute error: 139.77692772931726
 - Relative error: -0.30328821757393204
- (1, 4, 2)

- Value: 499.9453223388307
 - Absolute error: 139.7769277293171
 - Relative error: -0.21849627352249676
- (2, 1, 1)
 - Value: 657.9298829882995
 - Absolute error: 291.52093515955346
 - Relative error: -0.3070416387951929
- (2, 2, 1)
 - Value: 802.7501170117011
 - Absolute error: 291.5209351595538
 - Relative error: -0.26640651288464345
- (3, 5, 1)
 - Value: 628.8356548727498
 - Absolute error: 544.0586496133006
 - Relative error: -0.46385991263867654
- (4, 3, 1)
 - Value: 1876.1399999999994
 - Absolute error: 6.821210263296962e-13
 - Relative error: -3.635768259989639e-16
- (5, 4, 1)
 - Value: 628.6576444111025
 - Absolute error: 585.803975061562
 - Relative error: -0.4823569272744296

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 456.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 4, 2)
 - Value: 310.88996825396833
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 318.26999999999975
 - Absolute error: 160.21683881578912
 - Relative error: -0.3348406389030702

- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 317.51999999999987
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -3.5804622613257757e-16
- (5, 4, 1)
 - Value: 1111.1700317460316
 - Absolute error: 279.27322408783675
 - Relative error: -0.20085193906050677

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 685.55
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.6583303584219394e-16
- (1, 4, 2)
 - Value: 668.2999682539685
 - Absolute error: 279.273224087837
 - Relative error: 0.7178766711436555
- (2, 1, 1)
 - Value: 848.71999999999993
 - Absolute error: 160.21683881578963
 - Relative error: 0.23270312737594454
- (2, 2, 1)
 - Value: 1352.3699999999997
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.6812978359711624e-16
- (3, 5, 1)
 - Value: 2841.16
 - Absolute error: 4.547473508864641e-13

- Relative error: -1.6005693128386436e-16
- (4, 3, 1)
 - Value: 1857.2399999999993
 - Absolute error: 6.821210263296962e-13
 - Relative error: -3.672767258564839e-16
- (5, 4, 1)
 - Value: 2388.610031746031
 - Absolute error: 279.27322408783675
 - Relative error: 0.13239859233191334

$$(a) \ t^{(H)} - (1, 2)$$

$$(a) \ t^{(H)} - (1, 2)$$

$$(a) \ t^{(H)} - (2, 1)$$

$$(a) \ t^{(H)} - (2, 1)$$

$$(a) \ t^{(H)} - (3, 1)$$

$$(a) \ t^{(H)} - (4, 1)$$

$$(a) \ t^{(H)} - (5, 1)$$

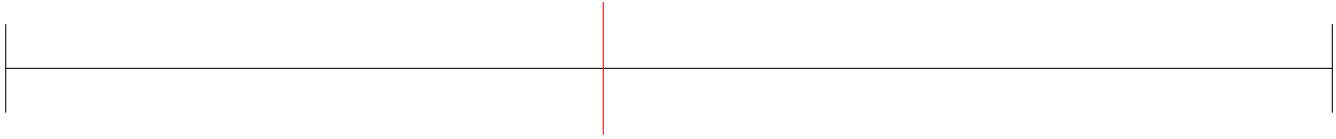
$$(a) \ t^{(H)} - (1, 1, 2)$$



$$(a) \ t^{(H)} - (1, 4, 2)$$



$$(a) \ t^{(H)} - (2, 1, 1)$$



$$(a) \ t^{(H)} - (2, 2, 1)$$



$$(a) \ t^{(H)} - (3, 5, 1)$$



$$(a) \ t^{(H)} - (4, 3, 1)$$



$$(a) \ t^{(H)} - (5, 4, 1)$$



$$(a) \ t^{(C)} - (1, 3)$$



$$(a) \ t^{(C)} - (4, 3)$$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 2)$



(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (3, 2)$



(a) $t^{(C)} - (4, 2)$



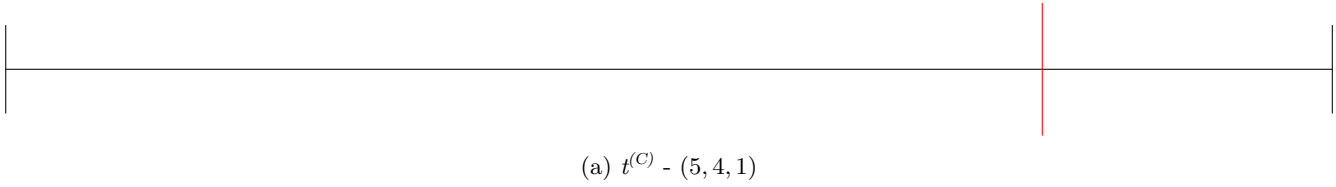
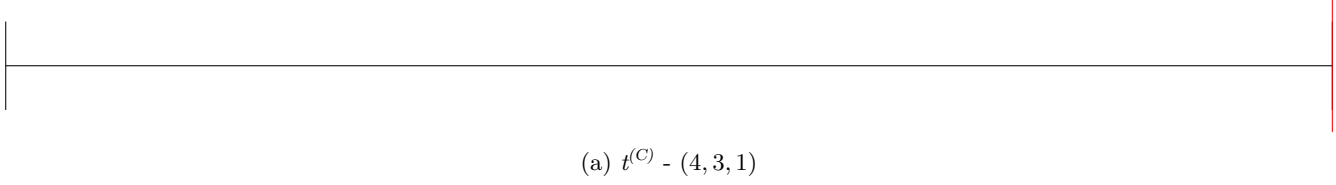
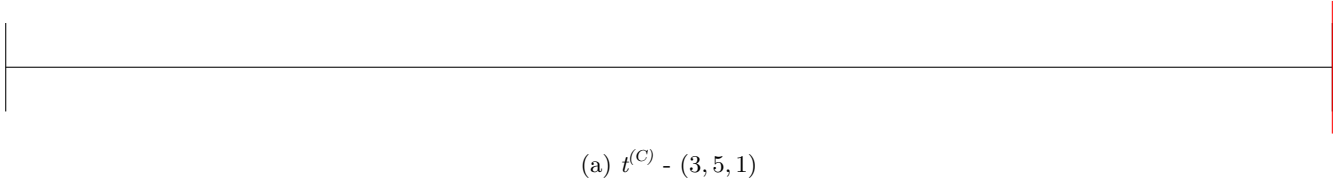
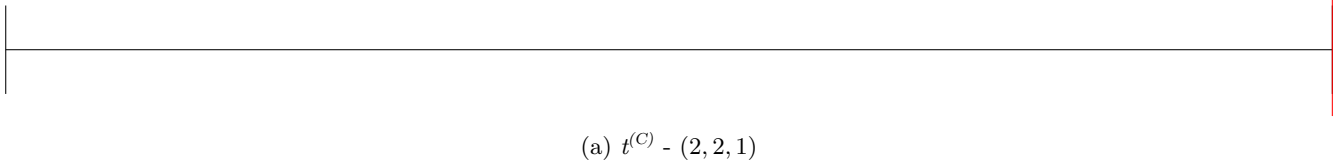
(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (1, 4, 2)$



(a) $t^{(C)} - (2, 1, 1)$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 2.700588
 - Error: 2.517749366732339
 - Relative error: 0.482481119227436
- $(1, 4, 2)$
 - Value: 5.404657
 - Error: 14.330675487747929
 - Relative error: 0.726143080629065
- $(2, 1, 1)$
 - Value: 3.303646
 - Error: 4.940238558717787
 - Relative error: 0.599260970349239
- $(2, 2, 1)$
 - Value: 18.890814
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(3, 5, 1)$
 - Value: 44.573765
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(4, 3, 1)$
 - Value: 19.921388
 - Error: 0.0000000000000050
 - Relative error: 0.0000000000000002
- $(5, 4, 1)$
 - Value: 12.878068
 - Error: 27.407136344457115
 - Relative error: 0.680327591322728

5.2 A_{CU}

- $cu, 3$
 - Value: 10.320104
 - Error: 10.559810021424973
 - Relative error: 0.505740107641329
- $cu, 5$

- Value: 14.682071
- Error: 13.381346891731138
- Relative error: 0.476825273318854



(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (1, 4, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (3, 5, 1)$



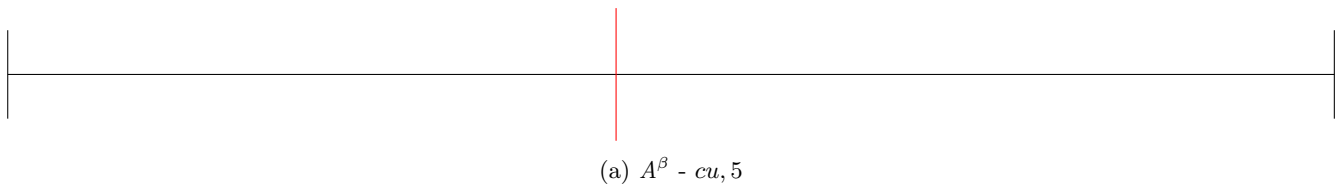
(a) $A^\beta - (4, 3, 1)$



(a) $A^\beta - (5, 4, 1)$



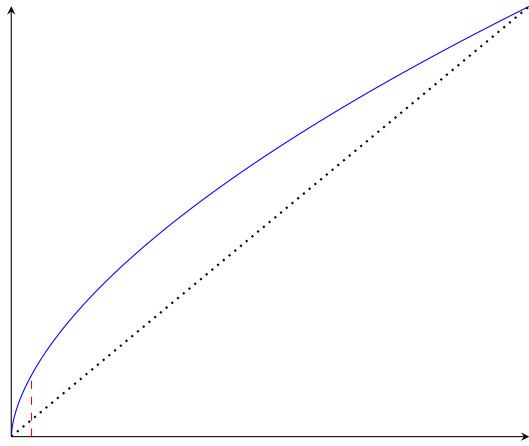
(a) $A^\beta - cu, 3$



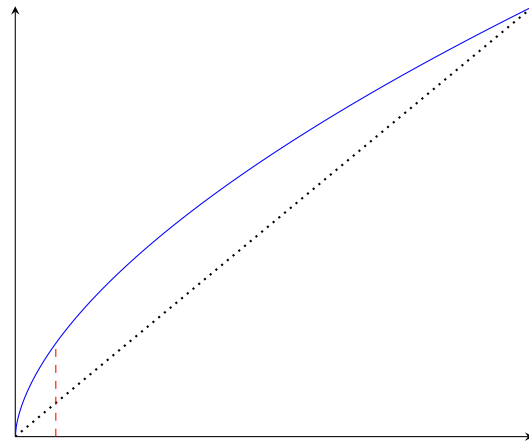
6 A^β Breakpoints

- (1, 1, 2)
 - q : 2.700588
 - $\hat{q}^\beta$: 0.496344
 - q^β : 1.814993
 - Breakpoints: [0, 69.05411764705883]
 - Error: 1.318649556927296
 - Relative error: 0.726531369078055
- (1, 4, 2)
 - q : 5.404657
 - $\hat{q}^\beta$: 0.993327
 - q^β : 2.752079
 - Breakpoints: [0, 69.05411764705883]
 - Error: 1.758751822678991
 - Relative error: 0.639062971503584
- (2, 1, 1)
 - q : 3.303646
 - $\hat{q}^\beta$: 0.547565
 - q^β : 2.048312
 - Breakpoints: [0, 89.41176470588236]
 - Error: 1.500747125526036
 - Relative error: 0.732675008665604
- (2, 2, 1)
 - q : 18.890814
 - $\hat{q}^\beta$: 3.338875
 - q^β : 5.831099
 - Breakpoints: [0, 76.14235294117645]
 - Error: 2.492223886234487
 - Relative error: 0.427402098982080
- (3, 5, 1)
 - q : 44.573765
 - $\hat{q}^\beta$: 5.563593
 - q^β : 9.759950
 - Breakpoints: [0, 181.68117647058827]
 - Error: 4.196357050275001
 - Relative error: 0.429956820490471
- (4, 3, 1)

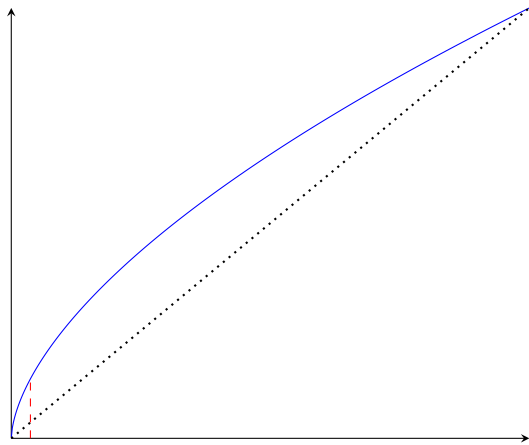
- q : 19.921388
- $\hat{q}^\beta$: 2.489490
- q^β : 6.019934
- Breakpoints: $[0, 181.14352941176475]$
- Error: 3.530443927227681
- Relative error: 0.586458871754244
- $(5, 4, 1)$
 - q : 12.878068
 - $\hat{q}^\beta$: 1.571184
 - q^β : 4.633514
 - Breakpoints: $[0, 192.3352941176471]$
 - Error: 3.062329236749322
 - Relative error: 0.660908660968983
- $cu, 3$
 - q : 10.320104
 - $\hat{q}^\beta$: 1.479788
 - q^β : 4.057051
 - Breakpoints: $[0, 128.44338870924773]$
 - Error: 2.577263237972494
 - Relative error: 0.635255350863578
- $cu, 5$
 - q : 14.682071
 - $\hat{q}^\beta$: 2.114093
 - q^β : 5.012708
 - Breakpoints: $[0, 127.1036185995295]$
 - Error: 2.898614506241985
 - Relative error: 0.578253233455313



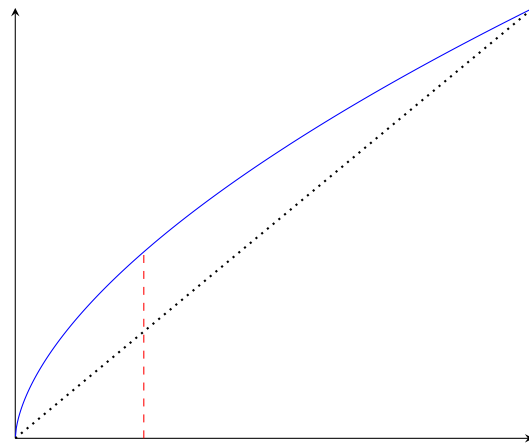
(a) $q^\beta - (1, 1, 2)$



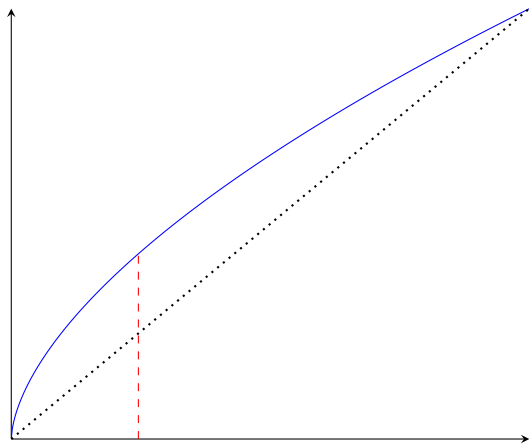
(b) $q^\beta - (1, 4, 2)$



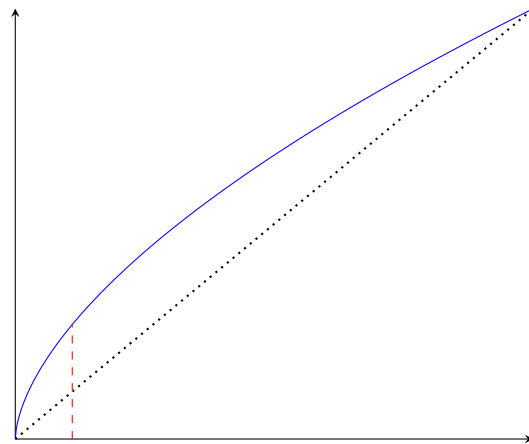
(c) $q^\beta - (2, 1, 1)$



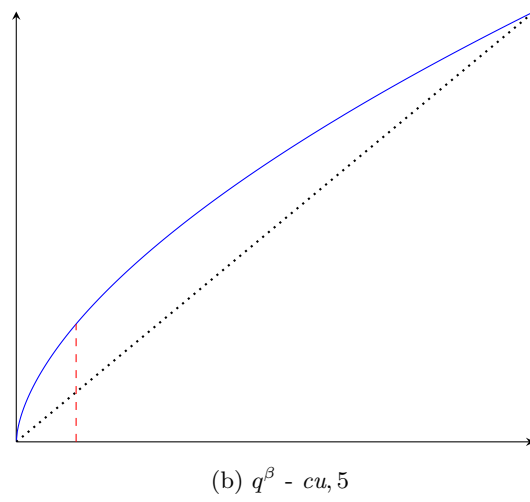
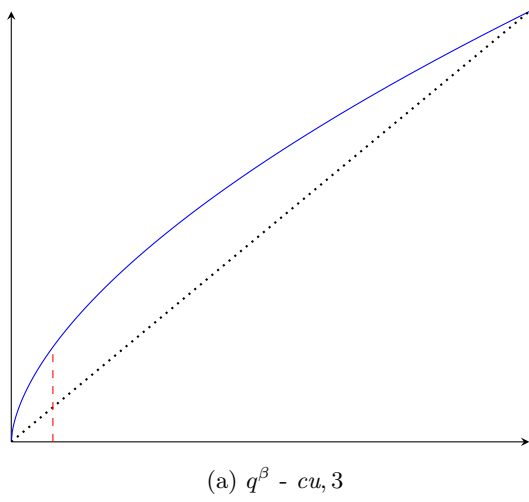
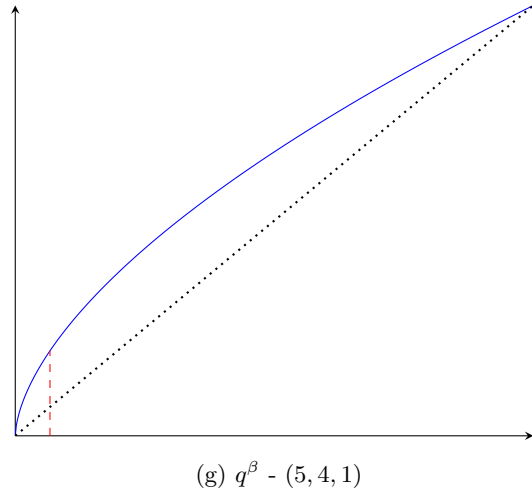
(d) $q^\beta - (2, 2, 1)$



(e) $q^\beta - (3, 5, 1)$



(f) $q^\beta - (4, 3, 1)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,2}$	-0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,4,2}$	-0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,2}$	0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{5,5,2}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,1}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{cu,2}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,2,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,2}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{2,2,3}$	106.1
$A_{hu,4}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{hu,5}^\beta$	0.0	$A_{4,5,1}$	0.0	$dt_{2,3,2}$	28.93737653834981
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,3}$	100.0
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,3}$	55.6
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,5,2}$	111.1
$A_{1,5,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,3}$	44.5
$A_{1,5,2}$	0.0	$A_{5,4,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,1,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,2,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,3}$	24.234831840647587
$A_{2,3,1}$	0.0	$A_{cu,1}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,3}$	22.64417509950809
$A_{2,4,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,3,2}$	10.0
$A_{2,5,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,3}$	27.8
$A_{3,1,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,4,1}$	50.0
$A_{3,1,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,4,2}$	130.5965435088267
$A_{3,2,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296989	$dt_{3,5,3}$	133.4
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	10.0	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296989	$dt_{4,1,2}$	10.0
$A_{3,4,1}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	88.9
$A_{3,4,2}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,2,1}$	122.2
$A_{3,5,2}$	0.0	$dt_{1,3,3}$	55.5	$dt_{4,2,2}$	10.0
$A_{4,1,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	33.3
$A_{4,1,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,3}$	183.3
$A_{4,2,1}$	0.0	$dt_{1,5,2}$	10.0	$dt_{4,4,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,3}$	66.7	$dt_{4,4,2}$	73.84046242774565
$A_{4,3,2}$	0.0	$dt_{2,1,3}$	100.0	$dt_{4,4,3}$	66.7

Variable	Value	Variable	Value	Variable	Value
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	10.0	$f_{2,3,1}^H$	0.0	$f_{1,1,1}^C$	2.2954999999999996
$dt_{4,5,3}$	55.6	$f_{2,3,2}^H$	10.6	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,4,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	23.552739491395734	$f_{2,5,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348401	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	3.782116402116406
$dt_{5,2,2}$	10.0	$f_{3,1,1}^H$	5.2140906269397895	$f_{1,5,1}^C$	0.0
$dt_{5,2,3}$	21.35698491649854	$f_{3,1,2}^H$	14.8	$f_{1,5,2}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,3,3}$	31.777163005386996	$f_{3,3,1}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,4,3}$	10.0	$f_{3,3,2}^H$	0.0	$f_{2,3,2}^C$	8.4
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,4,2}^C$	13.517883597883598
$dt_{5,5,3}$	10.0	$f_{3,5,2}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	0.0	$f_{2,5,2}^C$	13.9
$dt_{cu,2}$	97.8	$f_{4,1,2}^H$	12.6	$f_{3,1,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,2}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	8.116804201050268	$f_{3,5,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	17.7	$f_{4,1,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,5,1}^H$	8.8	$f_{5,3,2}^H$	0.0	$f_{4,3,2}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,4,2}^H$	0.0	$f_{4,4,1}^C$	0.0
$f_{2,1,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.010758450916196798	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.021193681833074414	$q_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.012093617925973806	$q_{1,5,1}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.01977531988328394	$q_{1,5,2}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.021502123357877267	$q_{2,1,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.021704019616484074	$q_{2,2,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,3,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,2}$	0.0064938037725891535	$q_{2,3,2}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.010199866868190046	$q_{2,4,1}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.013005645558983377	$q_{2,4,2}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,5,1}$	0.020971419097900186	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.025546702599208902	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.0209865189384691	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.0642496933351506	$\text{RecLMTD}_{5,1,2}$	0.03223630254302136	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.0	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.030706418976158206	$\text{RecLMTD}_{5,2,2}$	0.043215075417507326	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.0	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028825592541431463	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.03346772239731098	$\text{RecLMTD}_{5,4,2}$	0.035606497920977725	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.01264895322303948	$\text{RecLMTD}_{5,5,1}$	0.0	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.01616865595030577	$\text{RecLMTD}_{5,5,2}$	0.0	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.012362128482690376	$\text{RecLMTD}_{cu,1}$	0.01610890859262677	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.012937729640888148	$\text{RecLMTD}_{cu,2}$	0.009423141115769088	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.022197814115890363	$\text{RecLMTD}_{cu,4}$	0.008535688158765431	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.026432601068855872	$\text{RecLMTD}_{hu,1}$	0.012387051800747544	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.00826471540632166	$\text{RecLMTD}_{hu,2}$	0.04561830526586298	$q_{4,3,2}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.012806599920651586	$\text{RecLMTD}_{hu,3}$	0.05904512728898602	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.01838269073484035	$\text{RecLMTD}_{hu,4}$	0.01543447453012974	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.02882584079676209	$\text{RecLMTD}_{hu,5}$	0.024141568940966474	$q_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.03677912450245958	$q_{1,2,1}$	0.0	$q_{5,1,1}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0	$q_{5,1,2}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.02645080077831024	$q_{1,3,1}$	0.0	$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,2,2}^C$	115.6
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,3,1}^C$	37.8
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,3,2}^C$	37.8
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,4,1}^C$	82.2
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,4,2}^C$	82.2
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,5,1}^C$	93.3
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,2}^C$	93.3
$q_{cu,2}$	0.0	$t_{4,1,2}^H$	148.9	$t_{3,1,1}^C$	60.0
$q_{cu,4}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,2}^C$	60.0
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,2,1}^C$	147.8
$q_{hu,2}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	111.5
$q_{hu,4}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	82.2
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,4,2}^C$	82.2
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,5,2}^C$	93.3
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	126.72824858757069	$t_{4,1,1}^C$	60.0
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,2,1}^C$	115.6
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,3,2}^C$	37.8
$t_{1,5,2}^H$	160.0	$t_{5,4,2}^H$	65.6	$t_{4,4,1}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,2,2}^H$	137.8	$t_{5,5,2}^H$	65.6	$t_{4,5,1}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,2}^C$	93.3
$t_{2,3,2}^H$	137.8	$t_{1,2,1}^C$	147.8	$t_{5,1,1}^C$	60.0
$t_{2,4,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,4,2}^H$	137.8	$t_{1,3,1}^C$	111.5	$t_{5,2,1}^C$	147.8
$t_{2,5,1}^H$	248.9	$t_{1,3,2}^C$	37.8	$t_{5,2,2}^C$	115.6
$t_{2,5,2}^H$	137.8	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,1}^C$	111.5
$t_{3,1,1}^H$	226.7	$t_{1,5,1}^C$	102.2	$t_{5,3,2}^C$	37.8
$t_{3,1,2}^H$	122.35608108108104	$t_{1,5,2}^C$	93.3	$t_{5,4,2}^C$	82.2
$t_{3,2,1}^H$	226.7	$t_{2,1,2}^C$	60.0	$t_{5,5,1}^C$	102.2

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	93.3	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	1614.4323555888984	$b_{3,5,2}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	2243.0900000000001	$b_{4,1,1}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	1876.14
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	1408.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,2}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	1460.6800000000005	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	1614.4323555888986
$b_{2,4,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	2243.0900000000001
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	1182.0343451272502	$b_{1,5,1}^{H,out}$	1408.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	1810.8699999999997	$b_{1,5,2}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	367.27999999999994
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	1460.68	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	668.2999682539689
$b_{4,1,2}^{H,in}$	1876.14	$b_{3,1,1}^{H,out}$	1182.0343451272502	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1810.8699999999997	$b_{1,5,2}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,3,2}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,3,1}^{C,in}$	0.0
$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0	$b_{2,3,2}^{C,in}$	317.52

Variable	Value	Variable	Value	Variable	Value
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	1111.1700317460318	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	1296.87	$b_{1,4,1}^{C,out}$	668.2999682539689	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	317.52000000000055		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	1111.1700317460318		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	1296.87		
$b_{4,1,2}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,2}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	0.0		
$b_{5,4,2}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	367.2799999999994	$b_{4,5,2}^{C,out}$	0.0		
$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0		